

Lev Academic Center
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Reference Components Extraction from Hebrew-Aramaic Responsa

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Abstract

This thesis studies automatic extraction of reference components from Hebrew-Aramaic Jewish Responsa literature. The task is modeled as token-level named entity recognition (NER), where the model must identify authors, books, chapters, sections, and related prefix components inside citation-like text. The setting is difficult because the annotated corpus is small, the language is morphologically rich, the texts contain many abbreviations, and most tokens are not entities.

The final experimental corpus contains 300 annotated sentences, about 6,800 tokens, and 898 entity tokens. Approximately 86 percent of the tokens are labeled as non-entities. This makes the task a low-resource and highly imbalanced NER problem. The study compares four Hebrew transformer encoders: DictaBERT, BEREL 3.0, HeRo, and AlephBERT-Gimmel. It also compares three sentence-level train-test splitting strategies, targeted masked-language-model augmentation, and several NER inference pipelines: a regular one-stage NER model, a cascaded three-step model, confidence-based fusion, and SVM-based fusion.

All reported model comparisons are based on 20 seed-aligned runs. This design allows paired Wilcoxon signed-rank tests, rather than relying on a single split or a single training run. The strongest numerical mean F1 in the result workbook is BEREL 3.0 with baseline random split, augmentation, and SVM Fusion, reaching 0.857. However, because the random split is less stable for this dataset, the preferred robust configuration is BEREL 3.0 with Multilabel Stratified splitting, augmentation, and SVM Fusion, reaching 0.852. When SVM Fusion is treated as exploratory because its router was not trained on a separate held-out routing set, the strongest fully held-out non-router setting is BEREL 3.0 with Multilabel Stratified splitting, augmentation, and Confidence Fusion, reaching 0.848.

The main contribution is not a new transformer model. It is a controlled data-centric and model-centric framework for low-resource Hebrew-Aramaic reference extraction. The thesis shows that split strategy, augmentation, and fusion design can matter as much as model selection in small specialized corpora.

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Chapter 1

Introduction

Jewish Responsa literature is a large historical and legal corpus in which later authors cite earlier books, authors, chapters, and sections. These references are important for digital humanities, legal-historical research, and computational study of scholarly influence. A citation network can show which texts are central, which authorities are used by different communities, and how legal arguments develop over time (Satlow and Sperling, 2022; Ben-Gigi et al., 2024, 2025).

The computational task in this thesis is narrower than full citation-network construction. The goal is to identify the components that appear inside a reference. For example, a reference may contain an author name, a book name, a chapter marker, a chapter title, a section marker, and a section number or title. In this thesis the task is modeled as NER: each token receives a BIO label such as B-AUT, I-BOK, or O.

The problem is difficult for four reasons. First, the annotated dataset is small: only 300 sentences are available. Second, the class distribution is very imbalanced: the dominant label is O. Third, the text is Hebrew-Aramaic and includes morphology, prefixes, abbreviations, and spelling variation. Fourth, references are not always written in a fixed modern bibliographic format. They may be partial, abbreviated, or embedded in legal argumentation.

The research question is:

Can a combination of better data splitting, targeted augmentation, and structured or fused NER pipelines improve reference-component extraction in a small imbalanced Hebrew-Aramaic corpus?

The main contributions are:

1. A clear formulation of Responsa reference-component extraction as low-resource token-

level NER.

2. A comparison of four Hebrew transformer encoders under the same controlled experimental design.
3. A comparison of random, label-aware greedy, and multilabel stratified sentence splits.
4. A targeted augmentation method that increases rare entity representation only in the training set.
5. A comparison of regular NER, cascaded three-step prediction, confidence fusion, and SVM fusion.
6. A paired 20-seed evaluation with Wilcoxon signed-rank tests.

Chapter 2

Background and Related Work

2.1 Reference extraction in rabbinic and scholarly texts

Information extraction from scholarly documents often includes reference parsing, citation extraction, and linking extracted references to structured metadata. Large scholarly corpora such as S2ORC show the importance of citation and reference processing for downstream indexing and analysis (Lo et al., 2020). Similar ideas are relevant to historical religious corpora, although the language and reference formats are different.

In rabbinic literature, HaCohen-Kerner et al. (2010) showed that identifying biblical quotations is difficult because quotations may be free-form, unattributed, partial, or morphologically varied. Mughaz et al. (2014) and Moghaz et al. (2019) showed that extracting structured information from Hebrew, Aramaic, and Yiddish rabbinic texts can support historical inference, including the dating of authors and documents. Satlow and Sperling (2022) used rabbinic citations as a network of references and intellectual connections.

The closest recent work is Ben-Gigi et al. (2025), who constructed a citation network from medieval Jewish Responsa literature using automatic reference extraction. Their work is close to this thesis because both studies treat Responsa references as extractable structured information and use sequence-labeling ideas. The added value of this thesis is different: it focuses on a very small, highly imbalanced annotated setting and systematically studies data splitting, augmentation, multiple Hebrew encoders, cascaded prediction, and fusion under a paired multi-seed design. The thesis therefore provides methodological guidance for training reliable NER systems when only a few hundred labeled sentences are available. The results should not be interpreted as a direct score

comparison against Ben-Gigi et al. (2025), because the datasets and label schemas are not identical.

2.2 Hebrew language models

The models used in this thesis are all transformer encoders derived from the BERT or RoBERTa family (Devlin et al., 2019). AlephBERT showed the benefit of Hebrew-specific pretraining (Seker et al., 2022). AlephBERT-Gimmel extended this direction with a larger Hebrew vocabulary and improved Hebrew benchmarks (Gueta et al., 2023). HeRo introduced Hebrew RoBERTa and Longformer models (Shalumov and Haskey, 2023). DictaBERT was developed as a strong Modern Hebrew BERT suite (Shmidman et al., 2023). BEREL is especially relevant because it was trained for rabbinic-encoded language and is therefore closer to the Responsa domain (Shmidman et al., 2022).

A central expectation was that a domain-related model such as BEREL 3.0 could be more useful once the data preparation is strong enough. The results support this expectation in the strongest robust setting, where BEREL 3.0 performs best under Multilabel Stratified splitting with augmentation.

2.3 Low-resource and imbalanced NER

Modern NER is dominated by contextual embeddings and transformer-based models (Li et al., 2022; Hu et al., 2024; Keraghel et al., 2024). However, strong pretrained models do not remove the problems caused by small and imbalanced datasets. In imbalanced classification, random train-test splits can produce unstable estimates, especially when rare labels are missing from one side of the split (Japkowicz and Stephen, 2002). A careful evaluation design is therefore important (Kohavi, 1995; Pineau et al., 2021).

For NER, sequence structure also matters. CRF-based models explicitly encode transition constraints between neighboring labels (Lafferty et al., 2001; Ma and Hovy, 2016). This thesis does not train a CRF head, but it uses the same motivation when repairing invalid BIO sequences and enforcing consistency in the cascaded prediction pipeline.

2.4 Data augmentation, cascades, and fusion

Data augmentation is a common strategy when labeled data is limited. General text augmentation methods include simple lexical edits (Wei and Zou, 2019), while NER-

specific methods must be more careful because entity spans and labels must remain valid. MELM, proposed by Zhou et al. (2022), uses masked entity language modeling for low-resource NER. Recent work also studies artificial data generation with large language models for low-resource NER (Santoso et al., 2024). The augmentation in this thesis follows the NER-specific principle: generate new training examples while preserving the intended BIO structure and leaving the test set unchanged.

Cascaded and ensemble methods have long been used for NER and related classification tasks (Florian et al., 2003; Dinarelli and Rosset, 2011; Awasthy et al., 2020; Dietterich, 2000). Nguyen et al. (2023) showed that reformulating NER into easier low-resource subtasks can help under imbalance. This thesis uses a related decomposition idea, but the raw intermediate decomposition output is not reported as a final method. Instead, the reported cascaded method is the repaired three-step typed NER pipeline.

Chapter 3

Preprocessing and Dataset

3.1 Responsa text preprocessing

The early work in this thesis examined tokenization, abbreviation expansion, and POS tagging. These steps were useful because Responsa texts contain many abbreviations and punctuation patterns that are not handled well by a standard tokenizer.

A standard NLTK tokenizer (Bird et al., 2009) was tested, but it split Hebrew abbreviations into separate pieces and did not always split sentences at punctuation marks that are meaningful in Responsa texts. For example, colons, commas, and parentheses can mark useful reference boundaries. The final preprocessing therefore used a regular-expression tokenizer adapted to Hebrew Responsa text. It classified tokens into six broad groups: text tokens, regular abbreviations, shortened forms or acronym-like tokens, partial-stop punctuation, full-stop punctuation, and other punctuation.

Abbreviation expansion was provided by Dicta for tokens classified as abbreviations or shortened forms. A manual validation sample of 200 unique abbreviations, with up to 30 appearances per abbreviation, found that 72.77 percent of sampled expansions were correct, 19.37 percent were wrong, and 7.86 percent were not expanded. These preprocessing experiments informed the annotation and segmentation process, but not every preprocessing component was used as an input feature in the final NER experiments.

3.2 NER label schema

Each token is labeled using a BIO scheme. The entity types are shown in Table 3.1. Prefix categories such as pSEC and pCHP represent textual markers that introduce a section or chapter, while SEC and CHP represent the actual section or chapter component.

Table 3.1: Reference-component entity types.

Entity type	Meaning	Example role
AUT	Author	Name of cited author or sage
BOK	Book	Name of cited book or tractate
pSEC	Pre-section	Marker such as “section” or “siman”
SEC	Section	Section number or section name
pCHP	Pre-chapter	Marker such as “chapter” or “perek”
CHP	Chapter	Chapter name or number
pSCHP	Pre-sub-chapter	Marker introducing a sub-chapter component
SCHP	Sub-chapter	Sub-chapter number or title
0	Outside	Any non-entity token

3.3 Corpus size and entity distribution

The final annotated dataset contains 300 sentences, approximately 6,800 tokens, and 898 entity tokens. Table 3.2 shows the distribution of entity tokens before any split is applied. The largest entity type is AUT, while pSEC is very rare. This imbalance motivates the data-splitting and augmentation methods used later.

Table 3.2: Entity-token distribution in the annotated dataset.

Entity type	Tokens	Percent of entity tokens
AUT	308	34.30
BOK	180	20.04
CHP	120	13.36
SCHP	58	6.46
SEC	75	8.35
pCHP	81	9.02
pSCHP	54	6.01
pSEC	22	2.45
Total	898	100.00

Chapter 4

Methodology

4.1 Experimental design

Each experimental run is defined by four elements: a model, a method, a data condition, and a seed. The four models are DictaBERT, BEREL 3.0, HeRo, and AlephBERT-Gimmel. The reported methods are Regular NER, Cascaded 3-step, Confidence Fusion, and SVM Fusion. The data conditions are combinations of three split strategies with or without augmentation. Each condition is repeated over 20 random seeds.

Let a run be represented as

$$r = (m, a, c, s),$$

where m is the model, a is the method, c is the data condition, and s is the seed. For a fair paired comparison, the same seed list is reused across methods and models. This creates matched score vectors and enables paired significance testing.

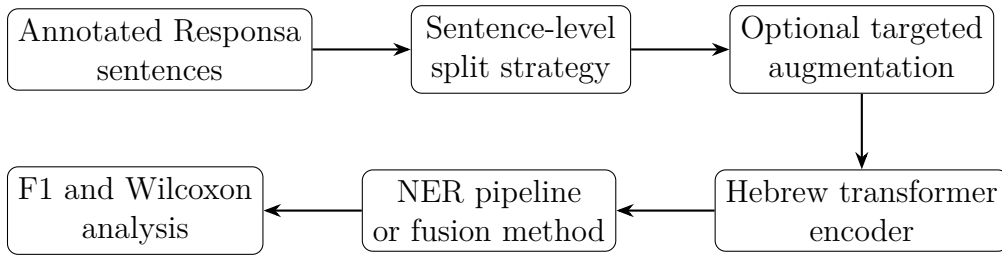


Figure 4.1: High-level experimental workflow.

4.2 Sentence-level split strategies

The dataset is split at sentence level. This avoids leakage where part of a sentence appears in training and another part appears in evaluation. Let the full dataset be

$$D = \{(x^{(j)}, y^{(j)})\}_{j=1}^N,$$

where each sentence $x^{(j)}$ has a sequence of token labels $y^{(j)}$. Each split creates disjoint sets

$$D_{train}^{(c,s)} \cap D_{eval}^{(c,s)} = \emptyset, \quad D_{train}^{(c,s)} \cup D_{eval}^{(c,s)} = D,$$

with an approximate 70/30 ratio.

4.2.1 Baseline random split

The baseline split shuffles the sentences with seed s and assigns about 70 percent to training and 30 percent to evaluation. This is simple, but it ignores label distribution. With a small imbalanced dataset, rare labels can become poorly represented in one of the splits.

4.2.2 Label-aware greedy split

The label-aware greedy split tries to improve entity coverage in the training set. For a subset A and label ℓ , define

$$count_A(\ell) = \sum_{(x,y) \in A} \sum_{i=1}^{|x|} \mathbf{1}[y_i = \ell].$$

The target training count is approximately

$$target_{train}(\ell) = 0.7 \cdot count_D(\ell).$$

The method greedily selects sentences that reduce the mismatch

$$\mathcal{L}_{split}(A) = \sum_{\ell \in \mathcal{Y} \setminus \{0\}} (count_A(\ell) - target_{train}(\ell))^2.$$

This method focuses mainly on improving the training set's coverage of non-0 labels.

4.2.3 Multilabel stratified split

The multilabel stratified method follows the logic of multilabel stratification (Sechidis et al., 2011; Szymanski and Kajdanowicz, 2017). Each sentence is treated as a multilabel item:

$$L_j = \{\ell : \ell \neq O, \ell \text{ appears in sentence } j\}.$$

For each label ℓ , the algorithm tracks how many sentences containing that label are still needed in each split. The sentence is assigned to the split with the larger remaining need. This strategy asks which split needs the sentence more in order to preserve the full label distribution.

4.3 Targeted masked-language-model augmentation

Augmentation is applied only to the training set. The evaluation set remains unchanged. This is important because otherwise the test target would also change.

For each entity label ℓ , let $f(\ell)$ be its frequency in the training set and let

$$f_{max} = \max_{\ell} f(\ell).$$

The deficit of label ℓ is

$$\Delta(\ell) = f_{max} - f(\ell).$$

The number of generated examples is proportional to this deficit:

$$g(\ell) = r \cdot \Delta(\ell),$$

with multiplier $r = 3$ in the final experiments.

The augmentation uses a masked replacement idea. Given a sentence

$$x = (w_1, \dots, w_k, \dots, w_n),$$

a relevant entity token is replaced by a mask:

$$x_{mask} = (w_1, \dots, [MASK], \dots, w_n).$$

The language model estimates

$$P(w \mid x_{mask})$$

and a replacement is selected from an entity-compatible vocabulary. The generated sen-

tence preserves the BIO span structure. This follows the general motivation of masked entity augmentation for low-resource NER (Zhou et al., 2022).

4.4 Regular one-stage NER

The regular NER model uses a transformer encoder followed by a token-classification head. For token i , the encoder produces a contextual vector h_i . A linear layer produces logits

$$z_i = Wh_i + b.$$

The probability of label k is

$$P(y_i = k \mid x) = \frac{\exp(z_{i,k})}{\sum_{k' \in \mathcal{Y}} \exp(z_{i,k'})}.$$

The predicted label is

$$\hat{y}_i = \arg \max_{k \in \mathcal{Y}} P(y_i = k \mid x).$$

The training objective is token-level cross-entropy over valid tokens:

$$\mathcal{L}_{NER} = - \sum_{i: y_i \neq -100} \log P(y_i \mid x).$$

4.5 Cascaded three-step NER

The cascaded method decomposes NER into three tasks: entity detection, BIO boundary prediction, and entity-type prediction. A full label is represented by:

$$e_i \in \{0, 1\}, \quad b_i \in \{B, I\}, \quad t_i \in \mathcal{T}.$$

For the 0 label, $e_i = 0$ and the boundary and type losses are ignored.

The model has one shared encoder and three heads:

$$z_i^{entity} = W_e h_i + b_e, \quad z_i^{bio} = W_b h_i + b_b, \quad z_i^{type} = W_t h_i + b_t.$$

Entity detection and boundary prediction use sigmoid probabilities, while type prediction uses a softmax over entity types. The final label is reconstructed as

$$\hat{y}_i^{cas} = \begin{cases} O, & \hat{e}_i = 0, \\ \hat{b}_i \hat{t}_i, & \hat{e}_i = 1. \end{cases}$$

The loss combines the three subtasks:

$$\mathcal{L}_{total} = \mathcal{L}_{entity} + \lambda_{bio}\mathcal{L}_{bio} + \lambda_{type}\mathcal{L}_{type}.$$

In the implementation, larger weights are assigned to boundary and type prediction because those losses are computed only on entity tokens.

After prediction, two structural repairs are applied. First, an I tag that appears after O or at the start of a sentence is changed to a B tag. Second, if a predicted B-X is followed by I-Y with $X \neq Y$, the inconsistent type is repaired using the stronger probability signal. This keeps the final BIO sequence structurally valid.

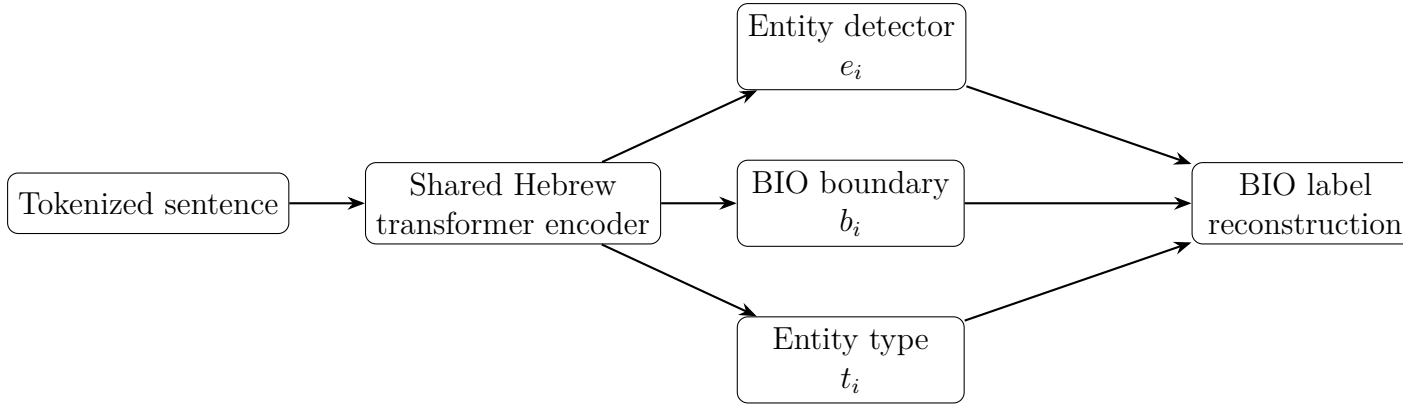


Figure 4.2: Cascaded three-step NER architecture.

4.6 Fusion methods

Confidence Fusion combines the regular model and the cascaded model. If both systems predict the same label, that label is retained. If they disagree, the label from the more confident system is selected. For token i :

$$\hat{y}_i^{fused} = \begin{cases} \hat{y}_i^{reg}, & \hat{y}_i^{reg} = \hat{y}_i^{cas}, \\ \hat{y}_i^{reg}, & \hat{y}_i^{reg} \neq \hat{y}_i^{cas} \text{ and } p_i^{reg} \geq p_i^{cas}, \\ \hat{y}_i^{cas}, & \text{otherwise.} \end{cases}$$

SVM Fusion is similar, but it does not use only the raw confidence comparison. It first collects disagreement tokens between the regular and cascaded systems. For training tokens where one of the two systems is correct, a linear SVM learns which source should be trusted. Features include confidence, entropy, probability margin, and disagreement information. The SVM then routes each disagreement token to either the regular or cascaded prediction.

A limitation is important here: in the present experiment, the SVM router is trained from the available ready predictions and is not evaluated on a separate held-out routing split. Therefore, SVM Fusion should be treated as an exploratory router or an upper-bound-style result, not as a fully independent final deployment result. The non-router Confidence Fusion results are more conservative.

4.7 Evaluation metrics and statistical tests

The primary metric is entity-level F1. Precision, recall, and F1 are defined as

$$Precision = \frac{TP}{TP + FP}, \quad Recall = \frac{TP}{TP + FN}, \quad F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}.$$

F1 is emphasized because accuracy can be misleading when most tokens are 0 (Sokolova and Lapalme, 2009; Chicco and Jurman, 2023).

For statistical testing, the study uses the paired Wilcoxon signed-rank test. For two methods A and B , scores are paired by seed:

$$d_s = F1_{A,s} - F1_{B,s}, \quad s \in \{1, \dots, 20\}.$$

The Wilcoxon test checks whether the median paired difference is different from zero. The significance level is $\alpha = 0.05$.

Chapter 5

Results

5.1 One-stage NER and split strategy

Table 5.1 shows the one-stage Regular NER results across split strategies. The random split is useful as a baseline, but it is not the most reliable split for this task. The label-aware and multilabel stratified splits generally improve the stability and coverage of entity labels. HeRo is strongest in the one-stage non-augmented setting, while BEREL 3.0 becomes stronger after augmentation in later tables.

Table 5.1: One-stage NER performance across sentence split strategies. Values are mean entity-level F1 $\pm$ standard deviation over 20 paired seeds.

Split strategy	DictaBERT	BEREL 3.0	HeRo	AlephBERT-Gimmel
Baseline random	0.672 ± 0.011	0.705 ± 0.001	0.745 ± 0.010	0.664 ± 0.001
Label-aware greedy	0.754 ± 0.000	0.707 ± 0.000	0.755 ± 0.000	0.658 ± 0.000
Multilabel stratified	0.738 ± 0.000	0.731 ± 0.006	0.769 ± 0.000	0.627 ± 0.004

5.2 Advanced pipelines without augmentation

Table 5.2 reports the advanced pipelines without augmentation. The main pattern is that SVM Fusion is the strongest reported method in most non-augmented settings. For example, DictaBERT reaches 0.811 under Label-aware greedy splitting with SVM Fusion, while HeRo reaches 0.805 under Multilabel Stratified splitting with SVM Fusion. However, without augmentation, BEREL 3.0 is not yet clearly dominant.

Table 5.2: Advanced pipeline performance without augmentation. Values are mean entity-level F1 $\pm$ standard deviation over 20 paired seeds.

Split strategy	Method	DictaBERT	BEREL 3.0	HeRo	AlephBERT-Gimmel
Baseline random	Regular NER	0.672 $\pm$ 0.011	0.705 $\pm$ 0.001	0.745 $\pm$ 0.010	0.664 $\pm$ 0.001
	Cascaded 3-step	0.737 $\pm$ 0.024	0.635 $\pm$ 0.110	0.739 $\pm$ 0.028	0.690 $\pm$ 0.035
	Confidence Fusion	0.686 $\pm$ 0.014	0.710 $\pm$ 0.013	0.753 $\pm$ 0.012	0.691 $\pm$ 0.020
	SVM Fusion	0.766 $\pm$ 0.016	0.771 $\pm$ 0.025	0.795 $\pm$ 0.013	0.739 $\pm$ 0.017
Label-aware greedy	Regular NER	0.754 $\pm$ 0.000	0.707 $\pm$ 0.000	0.755 $\pm$ 0.000	0.658 $\pm$ 0.000
	Cascaded 3-step	0.734 $\pm$ 0.028	0.652 $\pm$ 0.116	0.760 $\pm$ 0.019	0.723 $\pm$ 0.025
	Confidence Fusion	0.738 $\pm$ 0.011	0.715 $\pm$ 0.010	0.758 $\pm$ 0.011	0.710 $\pm$ 0.022
	SVM Fusion	0.811 $\pm$ 0.013	0.790 $\pm$ 0.028	0.799 $\pm$ 0.012	0.763 $\pm$ 0.011
Multilabel stratified	Regular NER	0.738 $\pm$ 0.000	0.731 $\pm$ 0.006	0.769 $\pm$ 0.000	0.627 $\pm$ 0.004
	Cascaded 3-step	0.764 $\pm$ 0.024	0.639 $\pm$ 0.147	0.750 $\pm$ 0.022	0.728 $\pm$ 0.031
	Confidence Fusion	0.745 $\pm$ 0.013	0.739 $\pm$ 0.012	0.776 $\pm$ 0.006	0.702 $\pm$ 0.020
	SVM Fusion	0.807 $\pm$ 0.010	0.795 $\pm$ 0.030	0.805 $\pm$ 0.011	0.766 $\pm$ 0.015

5.3 Advanced pipelines with augmentation

Table 5.3 shows the same methods after targeted augmentation. The augmented results are substantially higher for most model-method combinations. BEREL 3.0 becomes the strongest model in several settings. The highest numerical mean F1 in the reported results is 0.857 for BEREL 3.0 with baseline random split, augmentation, and SVM Fusion. The strongest robust controlled split setting is 0.852 for BEREL 3.0 with Multilabel Stratified splitting, augmentation, and SVM Fusion. Because the random split is known to be less stable for rare labels, the latter is the preferred main thesis result.

Table 5.3: Advanced pipeline performance with targeted augmentation. Values are mean entity-level F1 $\pm$ standard deviation over 20 paired seeds.

Split strategy	Method	DictaBERT	BEREL 3.0	HeRo	AlephBERT-Gimmel
Baseline random + augmentation	Regular NER	0.781 $\pm$ 0.011	0.836 $\pm$ 0.007	0.792 $\pm$ 0.011	0.781 $\pm$ 0.008
	Cascaded 3-step	0.790 $\pm$ 0.011	0.838 $\pm$ 0.010	0.796 $\pm$ 0.009	0.784 $\pm$ 0.011
	Confidence Fusion	0.788 $\pm$ 0.011	0.836 $\pm$ 0.009	0.785 $\pm$ 0.012	0.785 $\pm$ 0.013
	SVM Fusion	0.813 $\pm$ 0.012	0.857 $\pm$ 0.008	0.825 $\pm$ 0.015	0.809 $\pm$ 0.010
Label-aware greedy + augmentation	Regular NER	0.802 $\pm$ 0.011	0.824 $\pm$ 0.013	0.797 $\pm$ 0.015	0.813 $\pm$ 0.010
	Cascaded 3-step	0.805 $\pm$ 0.017	0.822 $\pm$ 0.007	0.785 $\pm$ 0.016	0.799 $\pm$ 0.011
	Confidence Fusion	0.807 $\pm$ 0.011	0.823 $\pm$ 0.011	0.787 $\pm$ 0.017	0.815 $\pm$ 0.012
	SVM Fusion	0.841 $\pm$ 0.013	0.844 $\pm$ 0.012	0.831 $\pm$ 0.013	0.841 $\pm$ 0.012
Multilabel stratified + augmentation	Regular NER	0.808 $\pm$ 0.010	0.847 $\pm$ 0.008	0.821 $\pm$ 0.009	0.834 $\pm$ 0.006
	Cascaded 3-step	0.828 $\pm$ 0.009	0.846 $\pm$ 0.008	0.825 $\pm$ 0.014	0.839 $\pm$ 0.010
	Confidence Fusion	0.814 $\pm$ 0.012	0.848 $\pm$ 0.008	0.823 $\pm$ 0.009	0.832 $\pm$ 0.008
	SVM Fusion	0.836 $\pm$ 0.008	0.852 $\pm$ 0.008	0.841 $\pm$ 0.010	0.847 $\pm$ 0.008

5.4 Effect of augmentation

Table 5.4 summarizes the augmentation effect by averaging paired F1 gains over the three split strategies. Augmentation improves all model-method combinations. The largest average gains occur for BEREL 3.0 and AlephBERT-Gimmel, suggesting that augmentation helps especially when the model has room to benefit from better rare-label coverage.

Table 5.4: Mean F1 gain from targeted augmentation, averaged over the three split strategies. Each value is the paired difference between augmented and non-augmented training for the same model, method, split, and seed.

Method	DictaBERT	BEREL 3.0	HeRo	AlephBERT-Gimmel
Regular NER	+0.075	+0.121	+0.047	+0.160
Cascaded 3-step	+0.062	+0.193	+0.052	+0.094
Confidence Fusion	+0.080	+0.114	+0.036	+0.109
SVM Fusion	+0.035	+0.066	+0.033	+0.076
Average	+0.063	+0.123	+0.042	+0.110

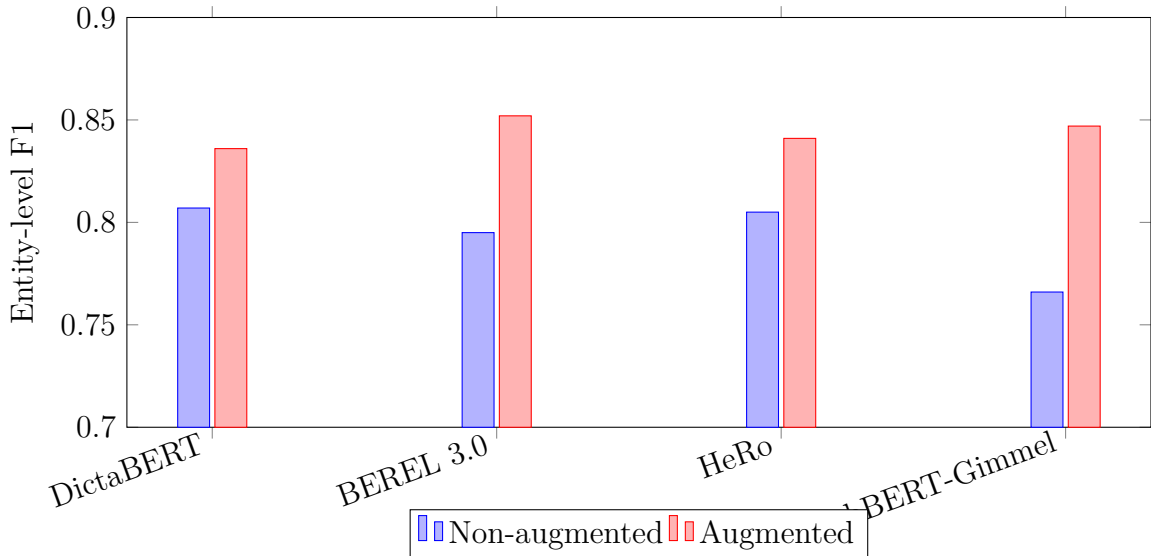


Figure 5.1: SVM Fusion under the Multilabel Stratified split, before and after augmentation.

5.5 Best configurations and significance tests

Table 5.5 shows the best mean F1 configuration for each model. All best configurations use augmentation and SVM Fusion. This supports the main conclusion that the largest improvements come from combining stronger data preparation with a fusion-based inference strategy.

Table 5.5: Best mean F1 configuration for each model among the reported methods.

Model	Best method	Split/data condition	F1
DictaBERT	SVM Fusion	Label-aware greedy + augmentation	0.841 ± 0.013
BEREL 3.0	SVM Fusion	Baseline random + augmentation	0.857 ± 0.008
HeRo	SVM Fusion	Multilabel stratified + augmentation	0.841 ± 0.010
AlephBERT-Gimmel	SVM Fusion	Multilabel stratified + augmentation	0.847 ± 0.008

Table 5.6 summarizes the Wilcoxon signed-rank tests. Augmentation improves all 48 paired comparisons and all are significant. Multilabel stratification is especially strong in the augmented setting. SVM Fusion significantly improves over Regular NER and Cascaded 3-step in all tested non-augmented and augmented split combinations, although the SVM router limitation should be remembered when interpreting these results.

Table 5.6: Summary of paired Wilcoxon signed-rank tests over seed-aligned runs. Positive means that the first item in the contrast improved F1.

Contrast	Setting	Positive	Significant	Mean $\Delta F1$	Largest p
Augmentation vs. same split without augmentation	All models, methods, splits	48/48	48/48	+0.085	3.9e-04
Label-aware greedy vs. baseline random	Non-augmented	14/16	10/16	+0.021	0.898
Multilabel stratified vs. baseline random	Non-augmented	15/16	13/16	+0.024	0.927
Label-aware greedy vs. baseline random	Augmented	11/16	13/16	+0.009	0.430
Multilabel stratified vs. baseline random	Augmented	15/16	16/16	+0.028	0.033
SVM Fusion vs. Regular NER	Non-augmented	12/12	12/12	+0.073	8.8e-05
SVM Fusion vs. Cascaded 3-step	Non-augmented	12/12	12/12	+0.071	2.6e-04
SVM Fusion vs. Confidence Fusion	Non-augmented	12/12	12/12	+0.057	9.5e-06
SVM Fusion vs. Regular NER	Augmented	12/12	12/12	+0.025	0.019
SVM Fusion vs. Cascaded 3-step	Augmented	12/12	12/12	+0.023	0.024
SVM Fusion vs. Confidence Fusion	Augmented	12/12	11/12	+0.025	0.076

5.6 Interpretation

The results support four main points. First, a single random split is not enough for a small imbalanced dataset. Even when a random split gives a high numerical score, it may not be the safest basis for conclusions. Second, targeted augmentation consistently improves performance because it increases exposure to rare entity patterns without changing the evaluation set. Third, model choice matters: BEREL 3.0 performs especially well once augmentation and a better split are used, which is consistent with its domain orientation. Fourth, fusion methods are useful because the regular and cascaded systems make different errors.

At the same time, the results should be interpreted carefully. The SVM Fusion method is the strongest reported method, but it is exploratory because the router was not trained

on a fully separate routing-development set. The Confidence Fusion result is therefore a more conservative estimate of the benefit of combining regular and cascaded predictions.

Chapter 6

Comparison with the Closest Work

The closest recent work is Ben-Gigi et al. (2025). Their study focuses on automatic construction of a citation network from medieval Jewish Responsa literature. It uses sequence labeling and rabbinic-domain BERT-style models to extract references at scale. This thesis shares the same broad goal: automatic extraction of structured references from Responsa literature.

The difference is the level of focus. Ben-Gigi et al. (2025) is primarily a large-scale citation-network construction study. This thesis is a controlled low-resource NER methodology study. It asks what should be done when the researcher has only a small annotated dataset, strong class imbalance, and several possible Hebrew encoders. Therefore, the contribution is not simply another extraction system. It is a detailed comparison of data splitting, augmentation, cascaded decomposition, fusion, and statistical testing.

The added value can be summarized as follows:

1. The thesis isolates the effect of sentence-level split design, showing that label-aware and multilabel splits can change conclusions in small data.
2. It evaluates four Hebrew encoders under the same seed-aligned framework.
3. It tests targeted augmentation for rare reference components.
4. It compares direct token classification with cascaded and fused prediction pipelines.
5. It reports paired Wilcoxon tests over 20 seeds rather than relying only on single-run scores.

This makes the work complementary to Ben-Gigi et al. (2025). A future study should apply the final method to a larger Responsa corpus and compare it directly against a BERT-CRF reference extractor under the same data and label definitions.

Chapter 7

Limitations and Future Work

7.1 Limitations

The main limitation is dataset size. With only 300 annotated sentences, even a 20-seed evaluation cannot fully replace a larger independent test set. A second limitation is that entity classes are strongly imbalanced, especially rare prefix classes such as **pSEC**. A third limitation is that the SVM Fusion router was not trained and evaluated using a fully separate routing split. A fourth limitation is that the experiments do not yet include a direct external comparison against the dataset and system of Ben-Gigi et al. (2025).

The augmentation method also has limits. Synthetic examples can help expose the model to rare labels, but they may also introduce unnatural examples if the replacement is semantically weak. For this reason, augmentation should be inspected manually in future work.

7.2 Future work

A first future direction is to enlarge the annotated corpus. This would make it possible to train a separate development set, test set, and routing set for the SVM Fusion method.

A second direction is to add explicit transition modeling, such as a CRF head or constrained decoding, rather than applying only post-hoc BIO repairs (Lafferty et al., 2001; Ma and Hovy, 2016).

A third direction is to add morphological analysis. Hebrew and Aramaic prefixes often carry important information about reference structure. Tools and methods from Hebrew NLP, including the direction discussed by Tsarfaty et al. (2019), could reduce segmentation errors.

A fourth direction is to combine transformer embeddings with stable handcrafted features. For example, quotation marks, abbreviations, prefix markers, and thesaurus matches for tractate or chapter names may help in low-resource conditions. This follows the broader finding that classical features can still be useful for small Hebrew-Aramaic tasks (HaCohen-Kerner et al., 2026).

Finally, the extraction system should be connected to downstream citation-network analysis and citation-intent classification. After references are extracted, one can ask whether a citation is supportive, critical, or neutral (Te et al., 2022). This would move the system from reference extraction toward interpretation of halachic authority and disagreement.

Chapter 8

Conclusion

This thesis studied reference-component extraction from Hebrew-Aramaic Responsa as a low-resource NER task. The main finding is that performance improves most when several components are combined: domain-relevant Hebrew encoders, careful sentence-level data splitting, targeted augmentation, and fused inference.

The best robust main result is BEREL 3.0 with Multilabel Stratified splitting, targeted augmentation, and SVM Fusion, reaching 0.852 mean F1 over 20 paired seeds. The highest numerical mean in the full result workbook is BEREL 3.0 with baseline random split, augmentation, and SVM Fusion, reaching 0.857, but this setting is less suitable as the main conclusion because random splitting is less reliable for rare labels. A conservative non-router result is BEREL 3.0 with Multilabel Stratified splitting, augmentation, and Confidence Fusion, reaching 0.848.

Overall, the thesis shows that in small specialized Hebrew-Aramaic corpora, the design of the data pipeline is not secondary to model selection. Better splitting, augmentation, and inference design can materially change the final NER result.

Appendix A

Suggested Figures to Add in the Final Submission

Figure placeholder. Add a per-label confusion matrix for the selected robust configuration: BEREL 3.0 + Multilabel Stratified + augmentation + SVM Fusion. This should show which entity types are confused with each other and which are confused with 0.

Figure placeholder. Add a per-label F1 bar chart for the selected robust configuration. This is important because the overall F1 may hide weak performance on rare labels such as pSEC.

Figure placeholder. Add one concrete Hebrew-Aramaic reference example with gold BIO labels and model predictions. This should help non-specialist readers understand what AUT, BOK, pCHP, CHP, pSEC, and SEC mean in real text.

Appendix B

Notes on Reproducibility

The experiments should be archived with the following items: the final annotated sentence file, the exact train-test split files for all seeds, the augmentation outputs, the model identifiers or local model checkpoints, and the full result workbook. The main reported comparisons require seed-aligned results for all models and methods.

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